VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

UNIT-01

Well posed learning problems

Well Posed Learning Problem - A computer program is said to learn from experience E in context to some task T and some performance measure P, if its performance on T, as was measured by P, upgrades with experience E.

Any problem can be segregated as well-posed learning problem if it has three traits -

- Task
- Performance Measure
- Experience

Certain examples that efficiently defines the well-posed **learning problem are -**

1. To better filter emails as spam or not

- Task - Classifying emails as spam or not
- Performance Measure - The fraction of emails accurately classified as spam or not spam
- Experience - Observing you label emails as spam or not spam

2. A checkers learning problem

- Task - Playing checkers game
- Performance Measure - percent of games won against opposer
- Experience **-** playing implementation games against itself

3. Handwriting Recognition Problem

- Task - Acknowledging handwritten words within portrayal
- Performance Measure - percent of words accurately classified
- Experience - a directory of handwritten words with given classifications

4. A Robot Driving Problem

- Task - driving on public four-lane highways using sight scanners
- Performance Measure - average distance progressed before a fallacy
- Experience - order of images and steering instructions noted down while observing a human driver

5. Fruit Prediction Problem

- Task - forecasting different fruits for recognition
- Performance Measure - able to predict maximum variety of fruits
- Experience - training machine with the largest datasets of fruits images

6. Face Recognition Problem

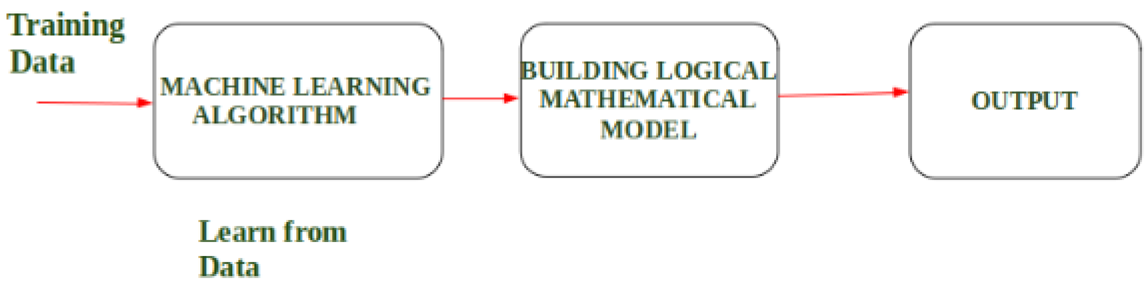
- Task - predicting different types of faces
- Performance Measure - able to predict maximum types of faces
- Experience - training machine with maximum amount of datasets of different face images

7. Automatic Translation of documents

- Task - translating one type of language used in a document to other language
- Performance Measure - able to convert one language to other efficiently
- Experience - training machine with a large dataset of different types of languages

Design a Learning System in Machine Learning

- According to Arthur Samuel "Machine Learning enables a Machine to Automatically learn from Data, improve performance from an Experience and predict things without explicitly programmed."
- In Simple Words, when we fed the Training Data to Machine Learning Algorithm, this algorithm will produce a mathematical model and with the help of the mathematical model, the machine will make a prediction and take a decision without being explicitly programmed.
- Also, during training data, the more machine will work with it the more it will get experience and the more efficient result is produced.



Example: In Driverless Car, the training data is fed to Algorithm like how to Drive Car in Highway, Busy and Narrow Street with factors like speed limit, parking, stop at signal etc.

After that, a Logical and Mathematical model is created on the basis of that and after that, the car will work according to the logical model. Also, the more data the data is fed the more efficient output is produced.

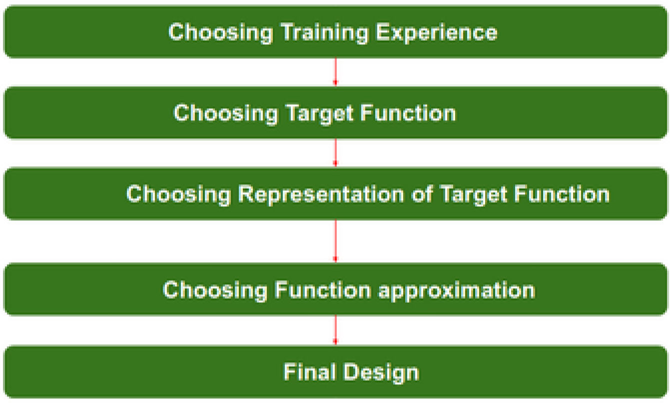
Designing a Learning System in Machine Learning:

According to Tom Mitchell, "A computer program is said to be learning from experience (E), with respect to some task (T). Thus, the performance measure (P) is the performance at task T, which is measured by P, and it improves with experience E."

Example: In Spam E-Mail detection,

- **Task, T:** To classify mails into Spam or Not Spam.
- **Performance measure, P:** Total percent of mails being correctly classified as being "Spam" or "Not Spam".
- **Experience, E:** Set of Mails with label "Spam"

Steps for Designing Learning System are:



Step 1) Choosing the Training Experience:

The very important and first task is to choose the training data or training experience which will be fed to the Machine Learning Algorithm. It is important to note that the data or experience that we fed to the algorithm must have a significant impact on the Success or Failure of the Model. So Training data or experience should be chosen wisely.

Below are the attributes which will impact on Success and Failure of Data:

- The training experience will be able to provide direct or indirect feedback regarding choices. For example: While Playing chess the training data will provide feedback to itself like instead of this move if this is chosen the chances of success increases.
- Second important attribute is the degree to which the learner will control the sequences of training examples. For example: when training data is fed to the machine then at that time accuracy is very less but when it gains experience while playing again and again with itself or opponent the machine algorithm will get feedback and control the chess game accordingly.

- Third important attribute is how it will represent the distribution of examples over which performance will be measured. For example, a Machine learning algorithm will get experience while going through a number of different cases and different examples. Thus, Machine Learning Algorithm will get more and more experience by passing through more and more examples and hence its performance will increase.

Step 2- Choosing target function: The next important step is choosing the target function. It means according to the knowledge fed to the algorithm the machine learning will choose NextMove function which will describe what type of legal moves should be taken. For example: While playing chess with the opponent, when opponent will play then the machine learning algorithm will decide what be the number of possible legal moves taken in order to get success.

Step 3- Choosing Representation for Target function: When the machine algorithm will know all the possible legal moves the next step is to choose the optimized move using any representation i.e. using linear Equations, Hierarchical Graph Representation, Tabular form etc. The NextMove function will move the Target move like out of these move which will provide more success rate. For Example: while playing chess machine have 4 possible moves, so the machine will choose that optimized move which will provide success to it.

Step 4- Choosing Function Approximation Algorithm: An optimized move cannot be chosen just with the training data. The training data had to go through with set of example and through these examples the training data will approximates which steps are chosen and after that machine will provide feedback on it. For Example: When a training data of Playing chess is fed to algorithm so at that time it is not machine algorithm will fail or get success and again from that failure or success it will measure while next move what step should be chosen and what is its success rate.

Step 5- Final Design: The final design is created at last when system goes from number of examples, failures and success, correct and incorrect decision and what will be the next step etc. Example: DeepBlue is an intelligent computer which is ML-based won chess game against the chess expert Garry Kasparov, and it became the first computer which had beaten a human chess expert.

Difference Between Supervised, Unsupervised and Reinforcement Learning.

Criteria	**Supervised Learning**	**Unsupervised Learning**	**Reinforcement Learning**
Input Data	Input data is labelled.	Input data is not labelled.	Input data is not predefined.
Problem	Learn pattern of inputs and their labels.	Divide data into classes.	Find the best reward between a start and an end state.

Solution	Finds a mapping equation on input data and its labels.	Finds similar features in input data to classify it into classes.	Maximizes reward by assessing the results of state-action pairs
Model Building	Model is built and trained prior to testing.	Model is built and trained prior to testing.	The model is trained and tested simultaneously.
Applications	Deal with regression and classification problems.	Deals with clustering and associative rule mining problems.	Deals with exploration and exploitation problems.
Algorithms Used	Decision trees, linear regression, K-nearest neighbors	K-means clustering, k-medoids clustering, agglomerative clustering	Q-learning, SARSA, Deep Q Network
Examples	Image detection, Population growth prediction	Customer segmentation, feature elicitation, targeted marketing, etc	Drive-less cars, self-navigating vacuum cleaners, etc

The differences between Supervised and Unsupervised learning are given below:

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in **Classification** and **Regression** problems.	Unsupervised Learning can be classified in **Clustering** and **Associations** problems.

Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorith